

# Theme 3.2 — Market Structures and the Labour Market

## 1. The Market Structure Spectrum

**Market structure** describes the competitive environment in which a firm operates. It is determined by: the number and size of firms, the degree of product differentiation, barriers to entry and exit, and the availability of information.

| Feature               | Perfect Competition | Monopolistic Competition | Oligopoly                     | Monopoly                      |
|-----------------------|---------------------|--------------------------|-------------------------------|-------------------------------|
| Number of firms       | Very many           | Many                     | Few dominant                  | One                           |
| Barriers to entry     | None                | Low                      | High                          | Very high                     |
| Product type          | Homogeneous         | Differentiated           | Differentiated or homogeneous | Unique (no close substitutes) |
| Price setting         | Price taker         | Some price setting       | Price maker (interdependent)  | Price maker                   |
| Non-price competition | None                | Branding, advertising    | Extensive                     | Limited (no rivals)           |

|                            |                  |                      |                           |                           |
|----------------------------|------------------|----------------------|---------------------------|---------------------------|
| SR supernormal profit      | Possible         | Possible             | Possible                  | Possible                  |
| LR supernormal profit      | No (normal only) | No (normal only)     | Yes (barriers protect it) | Yes (barriers protect it) |
| Allocative efficiency (LR) | Yes ( $P = MC$ ) | No ( $P > MC$ )      | No ( $P > MC$ )           | No ( $P > MC$ )           |
| Productive efficiency (LR) | Yes (min ATC)    | No (excess capacity) | Possible but unlikely     | No (not at min ATC)       |

Market structure is best thought of as a **spectrum** from perfect competition (most competitive) to pure monopoly (least competitive). Most real-world markets fall somewhere in between, with oligopoly and monopolistic competition being the most common.

## 2. Perfect Competition

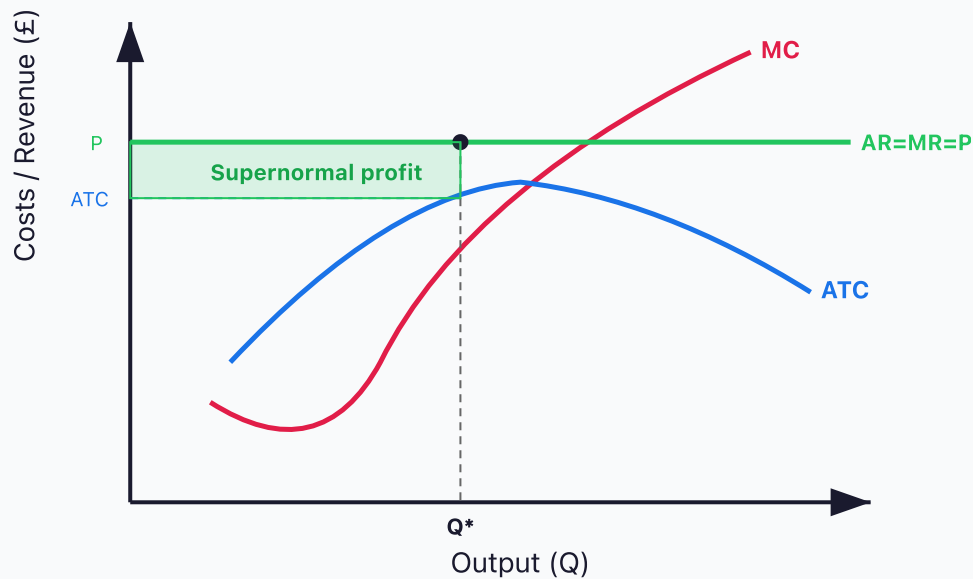
**Perfect competition** is a theoretical market structure with many buyers and sellers, homogeneous products, perfect information, no barriers to entry or exit, and each firm is a price taker facing a perfectly elastic demand curve.

### Assumptions (all must hold):

- Many small firms, each with a negligible market share
- Identical (homogeneous) products — consumers see no difference between firms
- Perfect knowledge for all buyers and sellers (of prices, costs, technology)
- Freedom of entry and exit — no barriers, no sunk costs
- Firms are **price takers** — they face a perfectly elastic (horizontal) demand curve at the market price
- No externalities, no transport costs, rational profit-maximising behaviour

Because the product is homogeneous, **AR = MR = Price** for each firm. If a firm tries to charge above the market price, it sells nothing (consumers switch to identical substitutes with perfect knowledge).

### Short-Run Equilibrium: Supernormal Profit



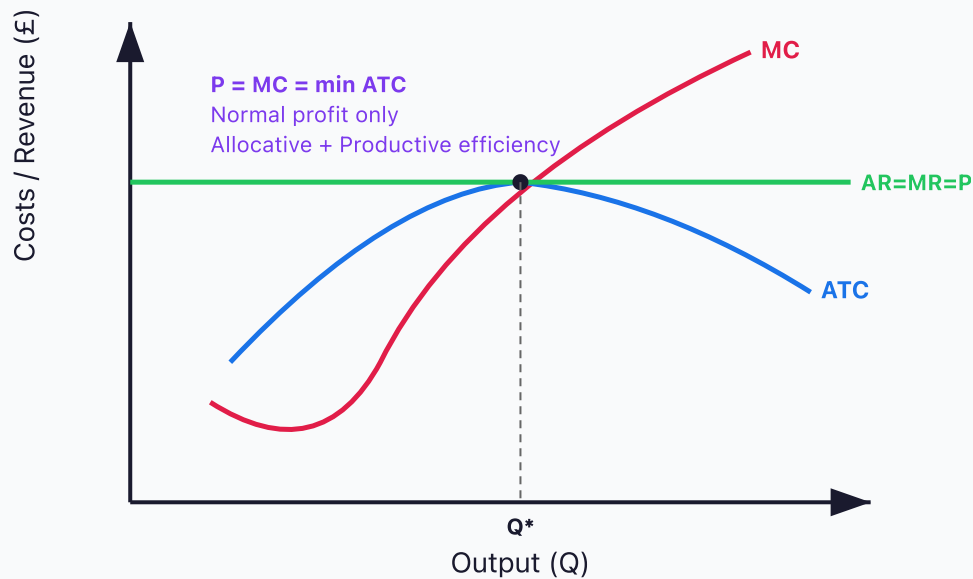
*Perfect competition: short-run supernormal profit.  $P > ATC$  at the profit-maximising output ( $MC = MR$ ).  
This attracts new entrants.*

## Long-Run Equilibrium: Normal Profit Only

### Adjustment mechanism:

1. Supernormal profits in the short run act as a **signal** to potential entrants
2. New firms enter the market (no barriers to entry)
3. Market supply increases → market price falls
4. This continues until price falls to the minimum of ATC
5. In long-run equilibrium:  **$P = MC = \text{minimum ATC}$**  — only normal profit is earned

The reverse occurs if firms make losses: firms exit → supply falls → price rises → until normal profit is restored.



Perfect competition long-run equilibrium:  $P = MC = \text{minimum ATC}$ . Allocative efficiency ( $P = MC$ ) and productive efficiency (minimum ATC) are both achieved. Only normal profit is earned.

## Efficiency in Perfect Competition

- **Allocative efficiency ( $P = MC$ ):** Resources are allocated to their most valued use. The price consumers pay equals the marginal cost of production.
- **Productive efficiency (min ATC):** Firms produce at the lowest possible average cost. No waste.
- **Dynamic efficiency:** Weak — normal profits may not fund R&D and innovation. This is the main criticism of perfect competition.
- **X-efficiency:** Achieved — competitive pressure eliminates managerial slack.

**Exam tip:** Perfect competition is a *benchmark model*, not a description of reality. Very few real-world markets satisfy all assumptions. Its value is in providing a standard against which to compare other market structures. The closest real-world examples are commodity markets (wheat, copper) and foreign exchange.

## 3. Monopolistic Competition

**Monopolistic competition** combines elements of monopoly (differentiated products, downward-sloping demand) and competition (many firms, low barriers to entry). Each firm has a small degree of market power due to product differentiation.

### Characteristics:

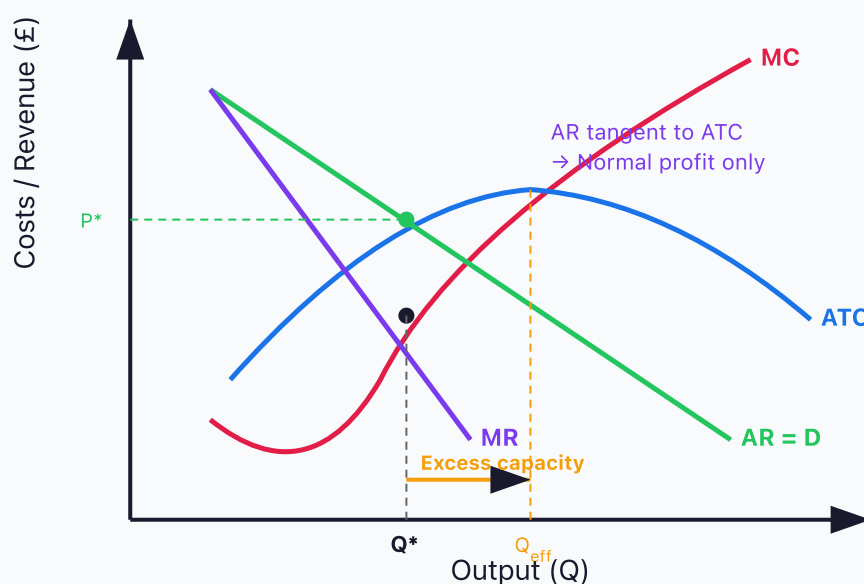
- Many firms, each with a small market share
- **Product differentiation** — products are similar but not identical (branding, quality, design, location)
- **Low barriers to entry and exit** — easy for new firms to enter
- Each firm faces a **downward-sloping demand curve** (some price-setting power) but demand is relatively elastic due to many substitutes
- **Non-price competition** — advertising, branding, customer service, product innovation

**Real-world examples:** Restaurants, hairdressers, clothing retailers, independent coffee shops.

### Short Run: Supernormal Profit Possible

In the short run, a monopolistically competitive firm can earn supernormal profit if demand is sufficiently high ( $AR > ATC$  at  $MC = MR$ ). The diagram is identical to the monopoly short-run diagram.

### Long Run: Normal Profit Only



*Monopolistic competition long-run equilibrium. AR is tangent to ATC (normal profit only). The firm produces at  $Q^*$  which is below the output at minimum ATC ( $Q_{eff}$ ), creating excess capacity.*

**Long-run adjustment:** Supernormal profits attract new entrants → each existing firm's demand curve shifts left (market share falls) → continues until AR is tangent to ATC → normal profit only.

## Efficiency in Monopolistic Competition

- **Allocative inefficiency:**  $P > MC$  — price exceeds marginal cost, so too little is produced from society's perspective
- **Productive inefficiency:** Firms produce to the left of minimum ATC — there is **excess capacity**. Firms could produce more at lower average cost but choose not to.
- **However:** Product differentiation gives consumers **variety and choice**, which has value. The "cost" of inefficiency may be offset by the benefit of diversity.

## 4. Oligopoly

**Oligopoly** is a market structure dominated by a **few large firms**, each with significant market share. The defining feature is **interdependence** — each firm's pricing and output decisions affect, and are affected by, the decisions of rival firms.

### Key characteristics:

- **High concentration ratio** — a few firms dominate (e.g. 5-firm concentration ratio  $> 50\%$ )
- **Interdependence** — firms must consider how rivals will react before making decisions
- **High barriers to entry** — economies of scale, brand loyalty, patents, sunk costs
- **Price rigidity** — prices tend to be stable (explained by the kinked demand curve)
- **Non-price competition** — advertising, loyalty cards, product innovation, quality, customer service
- Products may be **differentiated** (cars, smartphones) or **homogeneous** (oil, steel)

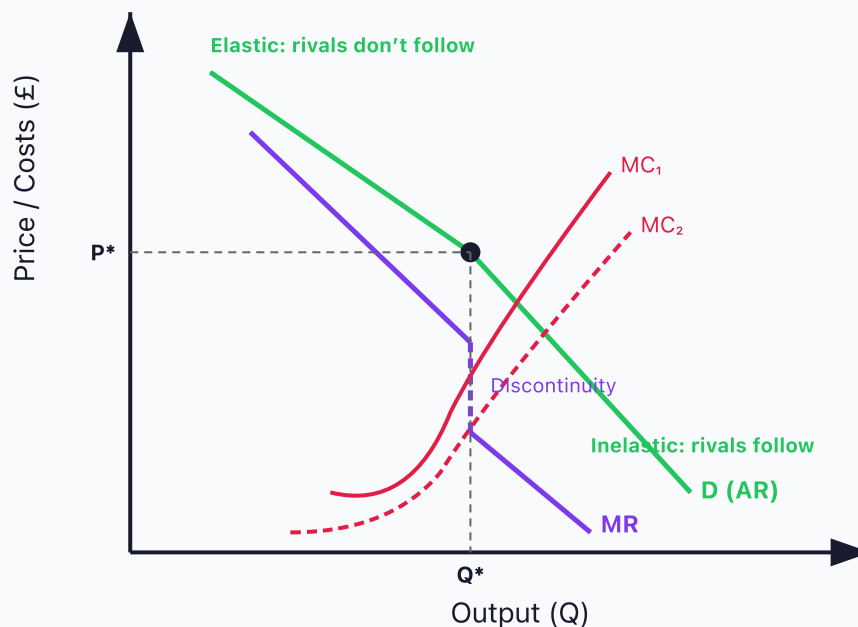
### 4a. The Kinked Demand Curve (Sweezy Model)

The **kinked demand curve** model explains price rigidity in oligopoly. It assumes:

- If a firm **raises** its price, rivals will *not* follow → the firm loses many customers → demand is

**elastic** above the current price

- If a firm **lowers** its price, rivals *will* follow (to protect market share) → the firm gains few extra customers → demand is **inelastic** below the current price



*The kinked demand curve. The kink at the current price creates a discontinuity (vertical gap) in the MR curve. MC can shift between  $MC_1$  and  $MC_2$  without changing the profit-maximising price or quantity, explaining price rigidity.*

**Key implication:** Because MC can shift within the vertical gap in MR without changing the profit-maximising output, prices tend to be **sticky** in oligopoly. Firms compete on non-price factors instead.

#### Evaluation of the kinked demand curve:

**Strengths:** Explains observed price rigidity; explains why non-price competition is common.

**Weaknesses:** Does not explain how the initial price is set; does not explain price wars when they do occur; assumes asymmetric rival responses which may not always hold; it is descriptive rather than predictive.

## 4b. Game Theory and the Prisoner's Dilemma

**Game theory** analyses strategic decision-making where the outcome for each player depends on the actions of all players. The **prisoner's dilemma** illustrates why oligopolistic firms may fail to collude even though collusion would maximise joint profits.

**Classic oligopoly example:** Two firms choose between a high price (cooperate) and a low price (compete).

|                    | Firm B: High Price | Firm B: Low Price |
|--------------------|--------------------|-------------------|
| Firm A: High Price | A: £5m, B: £5m     | A: £1m, B: £8m    |
| Firm A: Low Price  | A: £8m, B: £1m     | A: £3m, B: £3m    |

#### Analysis:

- The best joint outcome is both firms charging a high price (£5m each)
- But each firm has a **dominant strategy** to charge a low price: whatever the rival does, low price yields a higher payoff
- The **Nash equilibrium** is (Low, Low) — both firms charge a low price and earn £3m each
- This is a **suboptimal outcome** — both firms would be better off cooperating, but the incentive to cheat prevents it

In **repeated games** (which oligopoly resembles), firms can learn to cooperate through strategies like *tit-for-tat* (cooperate first, then mirror the rival's previous action). This explains why tacit collusion is more common than price wars in stable oligopolies.

## 4c. Collusion

**Collusion** occurs when firms in an oligopoly agree (formally or informally) to restrict competition, typically by fixing prices, limiting output, or dividing markets.

| Type                            | Description                                                                     | Legal Status        | Example         |
|---------------------------------|---------------------------------------------------------------------------------|---------------------|-----------------|
| <b>Overt (formal) collusion</b> | Explicit agreement between firms, often forming a cartel with quotas and prices | Illegal in UK/EU/US | OPEC oil cartel |

|                                   |                                                                                  |                              |                   |
|-----------------------------------|----------------------------------------------------------------------------------|------------------------------|-------------------|
| <b>Tacit (informal) collusion</b> | Firms follow a price leader without formal agreement; parallel pricing behaviour | Difficult to prove/prosecute | UK petrol pricing |
|-----------------------------------|----------------------------------------------------------------------------------|------------------------------|-------------------|

**Factors favouring collusion:** Few firms, similar cost structures, homogeneous product, high barriers to entry, stable demand, inelastic demand, weak regulation, repeated interaction.

**Factors against collusion:** Many firms, different costs, easy entry, volatile demand, strong regulation (CMA/EU Commission), incentive to cheat (prisoner's dilemma), differing objectives.

## 4d. Non-Price Competition

Because price competition can be destructive (price wars reduce all firms' profits), oligopolists often compete through:

- **Advertising and branding** — building brand loyalty and perceived quality
- **Product innovation and quality** — new features, improved design
- **Loyalty schemes** — Tesco Clubcard, Nectar points
- **Customer service** — delivery speed, after-sales support
- **Location and convenience**

### Case Study: UK Supermarket Oligopoly

The UK grocery market is a classic oligopoly: in 2024, Tesco (27%), Sainsbury's (15%), Asda (14%), and Morrisons (9%) together held roughly 65% market share (4-firm concentration ratio).

**Interdependence** is clear: when Aldi and Lidl grew rapidly in the 2010s, all major supermarkets responded with price-matching campaigns. **Non-price competition** is extensive: Tesco Clubcard, Sainsbury's Nectar, home delivery services, own-brand ranges, and convenience store formats. There have been instances of **tacit collusion** — in 2007, the OFT fined supermarkets for coordinating price rises on dairy products. The entry of discounters Aldi and Lidl illustrates how **contestability** can discipline incumbent oligopolists, though these markets still exhibit significant barriers (land, supply chains, brand loyalty).

## 5. Monopoly

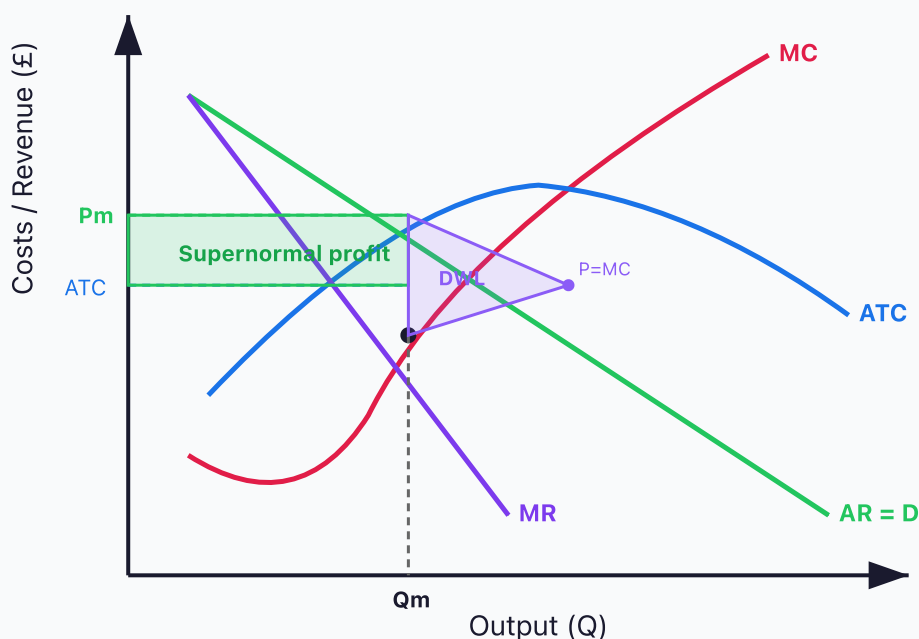
**Pure monopoly:** A single firm is the sole supplier in the market — it *is* the industry.

**Legal monopoly (UK):** A firm with 25%+ market share is considered to have monopoly power under the Enterprise Act 2002, allowing CMA investigation.

## Sources of Monopoly Power (Barriers to Entry)

- **Economies of scale / natural monopoly** — one firm can supply the whole market at lower cost than two or more firms (e.g. water pipes, rail network). The MES is close to or exceeds total market demand.
- **Legal barriers** — patents (up to 20 years), copyrights, licences, regulatory requirements
- **Brand loyalty and advertising** — huge marketing budgets create consumer loyalty that new entrants cannot match
- **Control of essential resources** — exclusive access to raw materials or technology
- **Predatory pricing** — temporarily setting price below cost to drive out competitors, then raising it once rivals have exited
- **High sunk costs** — irrecoverable costs that make exit costly, deterring entry
- **Network effects** — the value of a product increases with more users (e.g. social media, operating systems)

## Monopoly Diagram



Monopoly profit maximisation at  $MC = MR$ . The monopolist charges  $P_m$  (from  $AR$ ) and produces  $Q_m$ . Supernormal profit =  $(P_m - ATC) \times Q_m$ . The deadweight loss ( $DWL$ ) triangle shows the welfare loss compared to the competitive outcome ( $P = MC$ ).

## Advantages and Disadvantages of Monopoly

| Disadvantages (Costs)                                                           | Advantages (Benefits)                                                                                        |
|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Higher prices, lower output than competition                                    | <b>Economies of scale</b> may lead to lower costs and prices than many small firms                           |
| <b>Allocative inefficiency</b> ( $P > MC$ ) — underproduction                   | Supernormal profits fund <b>R&amp;D and innovation</b> (Schumpeter argument)                                 |
| <b>Productive inefficiency</b> (not at min ATC)                                 | <b>Natural monopoly</b> — duplication of infrastructure is wasteful (e.g. two sets of water pipes)           |
| <b>X-inefficiency</b> — lack of competition removes incentive to minimise costs | <b>Cross-subsidisation</b> — profits from one service fund loss-making services (e.g. rural postal delivery) |

|                                 |                                                                                             |
|---------------------------------|---------------------------------------------------------------------------------------------|
| Reduced <b>consumer surplus</b> | <b>International competitiveness</b> — large firms compete effectively in global markets    |
| <b>Deadweight welfare loss</b>  | <b>Dynamic efficiency</b> — monopoly profits provide the incentive and means for innovation |

### Schumpeter vs the neoclassical view:

The neoclassical view holds that monopoly is harmful:  $P > MC$ , deadweight loss, and X-inefficiency. However, **Schumpeter (1942)** argued that monopoly power is actually the engine of innovation. Only firms with supernormal profits can afford the risky R&D investments that drive "creative destruction" — new products and processes that benefit consumers in the long run. The truth likely depends on the specific market: in some industries (pharmaceuticals), patent-protected monopoly profits fund life-saving innovation; in others (cable TV), monopoly leads to poor service and high prices.

### Case Study: Google (Alphabet) — Search Engine Monopoly

Google holds approximately 90%+ of the global search engine market. Its monopoly power stems from **network effects** (more users → better data → better search results → more users), **brand loyalty**, and **economies of scale** in data centres. In 2024, a US federal court (in a case brought by the Department of Justice) ruled that Google maintained an illegal monopoly in search by paying Apple ~\$20bn/year to be the default search engine on iPhones. **Arguments for Google:** Free service for consumers, continuous innovation, dynamic efficiency. **Arguments against:** Dominance in digital advertising ( $P > MC$  for advertisers), stifles competition, raises barriers to entry for rival search engines, potential for data exploitation. This case shows the tension between static inefficiency and dynamic efficiency in monopoly evaluation.

## Price Discrimination

**Price discrimination** occurs when a firm charges **different prices to different consumers** for the same product, where the price difference is *not* justified by differences in cost.

### Three degrees:

- **1st degree (perfect):** Each consumer is charged the maximum they are willing to pay. All

consumer surplus is extracted. (e.g. auction, personal negotiation)

- **2nd degree:** Different prices for different quantities (e.g. bulk discounts, "buy 2 get 1 free")
- **3rd degree:** Different prices for different market segments based on identifiable characteristics (e.g. student discounts, peak/off-peak rail fares, age-based pricing)

### Conditions required:

- **Market power** — the firm must be a price maker (downward-sloping demand)
- **Ability to segment the market** — identify groups with different PED
- **Ability to prevent resale (arbitrage)** — consumers cannot buy cheaply and resell at a higher price
- **Different PED in each segment** — charge higher prices where demand is more inelastic

### Evaluation of price discrimination:

**Benefits:** Can increase output beyond single-price monopoly; some consumers who would be priced out can now buy (e.g. students); enables cross-subsidisation; may improve dynamic efficiency through higher profits funding R&D.

**Costs:** Reduces consumer surplus (especially 1st degree); can be seen as unfair/inequitable; those with inelastic demand (often the least well-off) pay the highest prices.

## 6. Contestable Markets

**Contestable markets** (Baumol, 1982) are markets where **potential competition** disciplines incumbent firms, even if the market is concentrated. The key condition is that **barriers to entry and exit are low** — particularly low **sunk costs** (costs that cannot be recovered on exit).

### Key features:

- **Freedom of entry and exit** — new firms can enter quickly and cheaply
- **Low sunk costs** — firms can exit without losing their investment
- Incumbent firms are disciplined by the **threat of hit-and-run entry**: if supernormal profits exist, new firms enter, take profits, and can exit costlessly if conditions change
- Even a monopolist in a perfectly contestable market must charge competitive prices to deter entry

**Implications:** The number of firms in a market matters less than the *threat* of entry. A monopoly in a contestable market may behave like a competitive firm — setting  $P$  close to  $AC$ , earning only normal

profit.

### Evaluation of contestability:

In practice, **perfectly contestable markets are rare**. Most markets have some sunk costs (advertising, specialised equipment, training). However, the theory is useful because it shifts focus from market structure to **barriers to entry** as the key determinant of firm behaviour. Examples of relatively contestable markets: budget airlines (aircraft can be leased and returned), some online markets (low setup costs), street food stalls.

## 7. Efficiency Comparisons Across Market Structures

**Allocative efficiency:**  $P = MC$  — resources are allocated to produce the goods consumers value most.

**Productive efficiency:** Production at minimum ATC — no wasted resources.

**Dynamic efficiency:** Firms innovate and improve products/processes over time, often funded by supernormal profits.

**X-efficiency:** Firms operate on (not above) their cost curves — no managerial slack.

| Efficiency Type         | Perfect Competition         | Monopolistic Competition        | Oligopoly                    | Monopoly                        |
|-------------------------|-----------------------------|---------------------------------|------------------------------|---------------------------------|
| Allocative ( $P = MC$ ) | Yes (LR)                    | No ( $P > MC$ )                 | No ( $P > MC$ )              | No ( $P > MC$ )                 |
| Productive (min ATC)    | Yes (LR)                    | No (excess capacity)            | Unlikely                     | No                              |
| Dynamic                 | Weak (no SN profit for R&D) | Moderate (some differentiation) | Strong (profits + incentive) | Potentially strong (Schumpeter) |
| X-efficiency            | Yes (competition forces it) | Yes (fairly competitive)        | Variable                     | No (no competitive pressure)    |

**Exam tip:** There is no single "best" market structure. Perfect competition achieves static efficiency but may lack dynamic efficiency. Monopoly can fund innovation but creates deadweight loss. The optimal outcome depends on the industry and the weight given to static vs dynamic efficiency. Always evaluate both sides.

## 8. The Labour Market

### 8a. Demand for Labour

**Demand for labour is a derived demand** — firms demand workers not for their own sake but because they produce goods and services that generate revenue. The level of labour demand depends on the demand for the final product.

**Marginal Revenue Product (MRP)** = Marginal Physical Product (MPP) × Marginal Revenue (MR)

MRP measures the *additional revenue* generated by hiring one more worker. The MRP curve is the firm's **demand curve for labour**.

$$\text{MRP} = \text{MPP} \times \text{MR}$$

Profit-maximising employment: hire workers until  $\text{MRP} = \text{MC of labour}$  (= wage in competitive market)

**Why does MRP slope downward?** Due to the **law of diminishing marginal returns**: as more workers are added to fixed capital, each additional worker's marginal product eventually falls, so MRP falls.

**Factors shifting the demand for labour (shifting MRP):**

- **Change in demand for the final product** — if demand for cars rises, demand for car workers rises (derived demand)
- **Change in productivity** — better technology or training increases MPP, shifting MRP right
- **Change in the price of the product** — higher product price increases MR, shifting MRP right

- **Change in price of substitute/complementary factors** — cheaper robots may reduce demand for manual labour (substitution effect) but also allow expansion (scale effect)

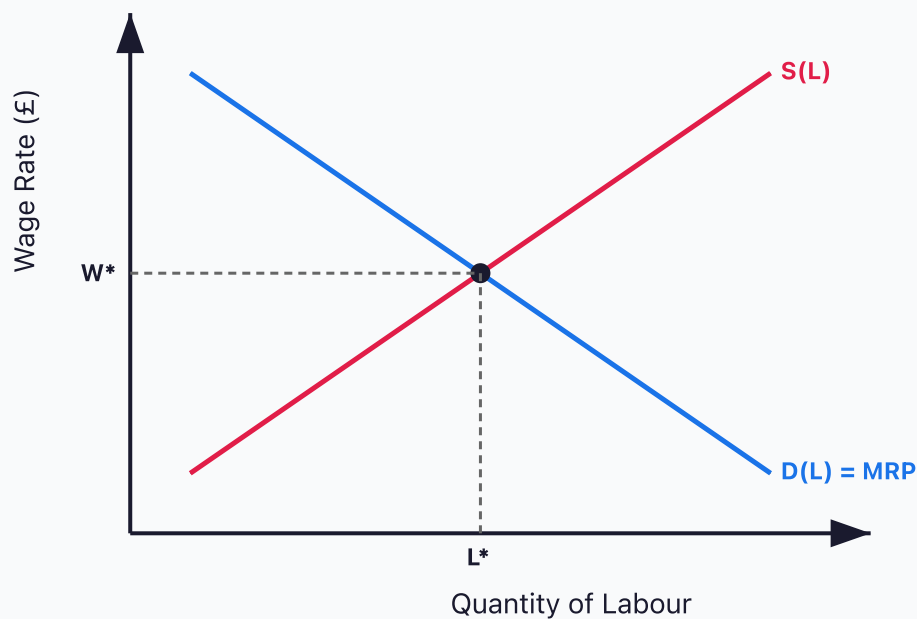
## 8b. Supply of Labour

**Individual labour supply:** At low wages, the **substitution effect** dominates (higher wage makes work more attractive relative to leisure → supply more hours). At high wages, the **income effect** may dominate (worker is wealthy enough to "buy" more leisure → supply fewer hours). This creates the **backward-bending supply curve** for individual labour.

**Market supply of labour to an occupation** is influenced by:

- **Wage rate** — higher wages attract more workers
- **Non-monetary factors** — job satisfaction, status, working conditions, job security, flexibility
- **Qualifications and training requirements** — high entry requirements restrict supply (e.g. surgeons)
- **Transfer earnings** — the wage available in the next best occupation
- **Net migration** — immigration increases labour supply
- **Population and demographics** — ageing population reduces labour supply
- **Benefits and tax system** — high benefits or tax may create a poverty/unemployment trap, reducing labour supply

## 8c. Wage Determination in a Competitive Labour Market



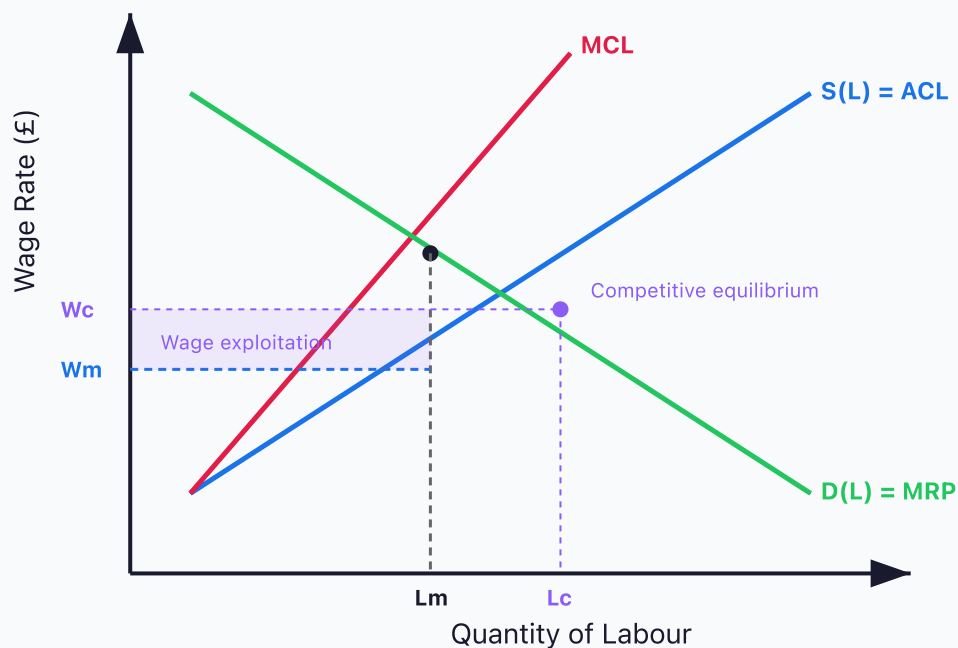
Competitive labour market equilibrium. The wage ( $W^*$ ) and employment level ( $L^*$ ) are determined by the intersection of labour demand (MRP) and labour supply.

### Factors causing wage differentials between occupations:

- **Differences in human capital** — education, skills, training, experience increase MRP
- **Compensating wage differentials** — dangerous, unpleasant, or unsociable work commands higher wages (e.g. oil rig workers, night shifts)
- **Differences in MRP** — some workers generate much more revenue (e.g. Premier League footballers)
- **Barriers to entry** — professional qualifications restrict supply (e.g. doctors, lawyers) → higher wages
- **Trade union power** — unions bargain for above-market wages
- **Monopsony power** — dominant employers suppress wages below competitive levels
- **Discrimination** — gender, race, age, disability
- **Geographical immobility** — workers may be unable to move to where wages are higher
- **Occupational immobility** — workers may lack skills to switch to higher-paying jobs

## 9. Monopsony in the Labour Market

**Monopsony** is a market with a **single (or dominant) buyer of labour**. The monopsonist is a wage-setter, not a wage-taker. Because it is the only employer, it faces an **upward-sloping supply curve** for labour: to hire more workers, it must raise the wage for *all* workers, not just the marginal one. Therefore, the **Marginal Cost of Labour (MCL) exceeds the wage (ACL)**.



*Monopsony: the firm hires where  $MCL = MRP$  (at  $L_m$ ) but pays the wage from the supply curve ( $W_m$ ). Compared to the competitive outcome ( $W_c, L_c$ ), monopsony results in lower wages and lower employment. The shaded area represents wage exploitation.*

**Key result:** The monopsonist hires where  $MCL = MRP$ , but pays the wage on the supply curve (ACL) at that employment level. This results in:

- **Lower wages** than in a competitive market ( $W_m < W_c$ )
- **Lower employment** than in a competitive market ( $L_m < L_c$ )
- **Exploitation** — workers are paid less than their MRP

**Real-world examples of monopsony:** NHS (dominant employer of nurses/doctors in UK), Amazon in warehouse towns, the military, school teachers (local authority is often the main employer).

**Exam tip:** Monopsony is one of the strongest justifications for a national minimum wage. If the labour market is monopsonistic, a minimum wage set between  $W_m$  and  $W_c$  can *increase both wages and employment* (moving towards the competitive outcome). This contradicts the simple supply-and-demand prediction that a minimum wage always reduces employment.

## 10. Trade Unions

**Trade unions** are organisations of workers that collectively bargain with employers over wages, working conditions, and terms of employment. They use their collective bargaining power to negotiate higher wages than individual workers could achieve alone.

### Types of union action:

- **Collective bargaining** — negotiating wages and conditions on behalf of members
- **Industrial action** — strikes, work-to-rule, overtime bans (used as a last resort)
- **Political lobbying** — influencing government policy on workers' rights, minimum wage, etc.

### Effects of trade unions:

- **Higher wages** — but may reduce employment if set above the competitive wage (creates a surplus of labour)
- **Improved working conditions** and job security
- **Counter monopsony power** — in a **bilateral monopoly** (monopsony employer vs union), the union can push wages closer to the competitive level, potentially *increasing* employment
- **Reduced wage inequality** within firms (union wage structures compress differentials)
- **Potential for wage-push inflation** if wage rises exceed productivity growth
- **May reduce labour market flexibility** — restrictive practices, resistance to new technology

### Evaluation — Are trade unions beneficial?

**Yes:** Correct monopsony exploitation, give workers a voice, improve safety, reduce inequality.

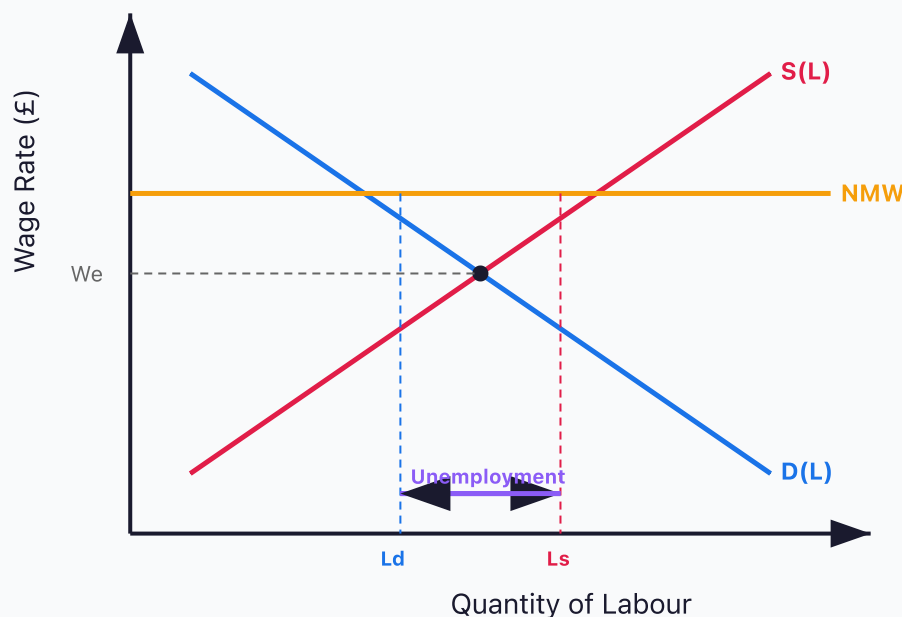
**No:** May cause unemployment (wage above equilibrium), reduce flexibility, contribute to inflation, benefit insiders (members) at the expense of outsiders (non-members who cannot find jobs at the union wage).

**Key point:** The impact depends on the existing market structure. In monopsonistic markets, unions are more likely to improve outcomes. In competitive markets, they may cause inefficiency.

## 11. National Minimum Wage and National Living Wage

The **National Minimum Wage (NMW)** is the legally mandated minimum hourly pay for workers, set by the government. The **National Living Wage (NLW)** is the higher rate for workers aged 21+ (as of 2024, £11.44/hour). The NLW is different from the **Real Living Wage** calculated by the Living Wage Foundation, which is voluntary and based on the actual cost of living.

### Effect of NMW in a Competitive Labour Market



*NMW set above the market equilibrium in a competitive labour market. Quantity of labour demanded ( $Ld$ ) is less than quantity supplied ( $Ls$ ), creating unemployment ( $Ls - Ld$ ).*

**In a competitive market:** A minimum wage above equilibrium causes a surplus of labour (unemployment). Firms hire fewer workers because the wage exceeds MRP for some.

**In a monopsony market:** A minimum wage between  $W_m$  and  $W_c$  can *increase* both wages and employment by effectively making the labour supply curve horizontal up to the NMW level, reducing

the monopsonist's incentive to restrict hiring.

### Case Study: The UK Minimum Wage Debate

The UK NMW was introduced in 1999 at £3.60/hour. Critics predicted mass unemployment, particularly among young and low-skilled workers. The evidence, however, has been more nuanced. The Low Pay Commission found that the NMW has **raised wages for the lowest-paid workers without significant negative employment effects**. This is consistent with the **monopsony model** — many low-paid workers are in sectors (care homes, hospitality, retail) where employers have significant wage-setting power. However, there are concerns that very large increases (e.g. the NLW rising to £11.44 in 2024) could start to affect employment in some sectors, particularly small businesses with thin margins. The debate illustrates the importance of **elasticity of demand for labour**: if demand is inelastic, employment effects are small; if elastic, job losses may be significant.

## 12. Discrimination in the Labour Market

**Labour market discrimination** occurs when workers are treated differently (in hiring, pay, or promotion) based on characteristics **unrelated to productivity**, such as gender, race, age, disability, or religion.

### Types of discrimination:

- **Direct discrimination** — explicitly treating someone less favourably (e.g. refusing to hire women)
- **Indirect discrimination** — applying rules that disproportionately disadvantage a group (e.g. height requirements that exclude women)
- **Statistical discrimination** — using group averages to make decisions about individuals (e.g. assuming young workers will leave for childcare)
- **Occupational segregation** — certain groups concentrated in lower-paying jobs ("sticky floor" / "glass ceiling")

### Consequences of discrimination:

- **Allocative inefficiency** — workers are not matched to jobs based on productivity, misallocating resources
- **Lower output** — talented workers are underemployed or excluded, reducing GDP

- **Wage inequality** — discriminated groups earn less than their MRP
- **Social costs** — poverty, reduced social mobility, inequality of opportunity

#### Government policies to address discrimination:

- **Equal Pay Act (1970) / Equality Act (2010)** — illegal to pay differently for same work based on protected characteristics
- **Gender pay gap reporting** — firms with 250+ employees must publish pay gap data (since 2017)
- **Positive action** — encouraging underrepresented groups to apply (legal) vs positive discrimination (illegal in UK except disability)
- **Education and training** — addressing occupational segregation through broadening opportunities

#### Case Study: The UK Gender Pay Gap

In 2023, the UK gender pay gap (median) was approximately 14.3% for all employees (full and part-time). For full-time employees only, it was around 7.7%. **Causes include:** occupational segregation (women concentrated in lower-paid caring/service roles), the "motherhood penalty" (women taking career breaks for childcare), unconscious bias in hiring and promotion, and undervaluation of work in female-dominated sectors. **Policy responses:** The Equality Act 2010, mandatory gender pay gap reporting (introduced 2017), shared parental leave (2015), and free childcare for 3–4 year olds. **Evaluation:** The pay gap has narrowed significantly since the 1970s but progress has slowed. Pay gap reporting increases transparency but does not directly mandate change. Addressing the root causes (occupational segregation, caring responsibilities) requires deeper structural reform.

## Key Formulae and Conditions Summary

| Concept               | Formula / Condition                        |
|-----------------------|--------------------------------------------|
| MRP                   | $MPP \times MR$                            |
| Profit-max employment | $MRP = MCL$ (= wage in competitive market) |
| Allocative efficiency | $P = MC$                                   |
| Productive efficiency | Production at minimum ATC                  |

|                             |                                                              |
|-----------------------------|--------------------------------------------------------------|
| Normal profit               | $TR = TC$ ( $AR = ATC$ )                                     |
| Supernormal profit          | $TR > TC$ ( $AR > ATC$ )                                     |
| Perfect competition LR      | $P = MC = \min ATC$ (normal profit)                          |
| Monopolistic competition LR | AR tangent to ATC (normal profit, $P > MC$ )                 |
| Monopoly / Oligopoly        | $MC = MR$ , P from AR curve (supernormal profit possible LR) |
| Monopsony employment        | $MCL = MRP$ , wage from $S(L)$                               |

### Exam tip — 25-mark essay structure for market structure questions:

1. Define the market structure and its key features.
2. Draw and explain the relevant diagram (show price, output, profit).
3. Analyse efficiency outcomes (allocative, productive, dynamic, X-efficiency).
4. Evaluate with real-world examples — does the theoretical model match reality?
5. Consider contestability — how do barriers to entry affect the outcome?
6. Conclude: the best market structure depends on the criteria used (static vs dynamic efficiency, consumer vs producer welfare).